

# Islam Kathat

AI Engineer | AI Agents & RAG Systems

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## SUMMARY

AI engineer focused on the infrastructure behind AI agents and RAG systems. Pretrained a 350M-parameter GPT end to end on 6.83B tokens – data pipeline, remote GPU training loop, profiling, crash recovery – then ran SFT and QLoRA post-training on top. Moves between low-level PyTorch work and the tooling that keeps other people's runs fast and reliable.

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL, TypeScript, JavaScript, Bash

**ML & Training:** PyTorch, torch.compile, CUDA, Transformers, TRL, PEFT, Accelerate, BitsAndBytes, Flash Attention, mixed precision (BF16/FP16), gradient checkpointing, distributed and multi-GPU training, supervised fine-tuning, LoRA/QLoRA, NF4 quantization, KV cache, NumPy, Pandas, Scikit-learn

**Infrastructure & Tooling:** Docker, Redis, RabbitMQ, Kafka, AWS S3, Weights & Biases, Hugging Face Hub, RunPod, rclone, Git, pytest, Linux, profiling and performance debugging, FastAPI, PostgreSQL, MongoDB, BM25 and embedding hybrid retrieval, React, Next.js

## EXPERIENCE

### JSB Global Infotech

May 2025 – May 2026

Full-Stack AI Engineer

Jaipur, India

- Built and operated backend services on FastAPI, Node.js, Redis, RabbitMQ, Docker, and AWS S3 running scheduled AI pipelines that scraped, rewrote, and published content with no manual steps.
- Fine-tuned and deployed a retrieval-grounded LLaMA 3.1 support assistant on the company site, replacing templated replies with answers grounded in the company's own documents.
- Cut API response time 35% by rewriting MongoDB aggregation pipelines and adding Redis caching across React Native (Expo) and Next.js products serving live users.

### CodeAlpha

Jul 2024 – Aug 2024

Data Science Intern

Remote

- Rebuilt feature engineering and hyperparameter search for classification tasks in Scikit-learn and PyTorch, raising held-out accuracy 15% over the initial pipeline.

## PROJECTS

### 350M-Parameter GPT: Pretraining, Post-Training & Training Infrastructure

[GitHub](#) | [Model](#) | [Training logs](#)

PyTorch, CUDA, torch.compile, Flash Attention, BF16, Hugging Face, Weights & Biases

- Trained a decoder-only transformer (24 layers, 16 heads, 1024 dim, 50,304 vocab, RMSNorm, Flash Attention, weight tying) on 6.83B tokens to validation loss 3.04 and perplexity 20.9, comparable to GPT-2 XL at 6x fewer tokens.
- Diagnosed a recurring torch.compile and BF16 out-of-memory failure by tracing it to AOT Autograd materializing 12GB FP32 gradient buffers, then excluded lm\_head and cross\_entropy from the compiled graph via torch.compiler.disable to restore training on the same hardware.
- Built a fault-tolerant data and checkpointing pipeline: streamed and tokenized the corpus on free Colab CPU into uint16 shards, shipped them to rented RunPod GPUs with rclone, and made runs resumable from the last checkpoint.
- Ran supervised fine-tuning on OpenHermes 2.5 (746K examples, 288M tokens) with identity-protection examples to prevent factual bleeding; tracked every run in Weights & Biases and shipped to Hugging Face.

### ParcelPilot AI Support Agent: Tool-Calling Agent & Verification Harness

[Live](#) | [Backend](#) | [Frontend](#)

Python, FastAPI, streaming tool-calling loop, hybrid retrieval, pytest, React, TypeScript

- Wrote the tool-calling loop rather than adopting a framework: 7 tools, a bounded 6-step budget, per-index accumulation of fragmented argument deltas before dispatch, and every call, result, and token streamed live.
- Enforced authorization in the tool layer, not the prompt: the caller's principal is injected server-side and never exposed as a model-writable argument, so cross-account reads fail even when the model is told to.
- Pinned agent behaviour with 18 pytest cases covering cross-account isolation, source precedence, and a confirm-before-execute gate proving state-changing tools write nothing until confirmed.

### Mistral-7B QLoRA Fine-Tuning

[GitHub](#) | [Model](#) | [Training logs](#)

Adapted Mistral-7B-Instruct-v0.3 on 500 domain instruction pairs by training 0.5% of parameters under NF4 4-bit quantization (r=64), cutting perplexity 13% with a cosine LR schedule, paged AdamW, and gradient checkpointing.

**Additional:** [RAG system](#) (FastAPI, ChromaDB, LLaMA 3.1, cross-encoder re-ranking; live) | [CIFAR-10 benchmark](#) (92-97%)

## EDUCATION

JECRC University – MCA, Computer Science, CGPA 8.57/10

Jul 2025 | Jaipur, India

MDS University – BCA, Computer Science, CGPA 7.68/10

Jul 2023 | Ajmer, India

**Certifications:** [Claude 101](#) and [AI Fluency \(Anthropic\)](#) | [Responsible Generative AI \(Microsoft\)](#) | [Deep Learning \(KPMG\)](#)